

Table 1 Measured boiling incipience superheats for all surfaces and fluids tested, along with calculated incipient bubble radii. Values in table are the maximum single-phase wall superheats measured and accuracy is based upon the magnitude of the input power increment

Surface	HFE-7300 ^a		DI water ^b	
	$\Delta T_{w,i}$ (K)	$r_{b,i}$ (μm)	$\Delta T_{w,i}$ (K)	$r_{b,i}$ (μm)
Bare copper	22.5 ± 2.0	0.279 ± 0.026	9.7 ± 0.6	3.51 ± 0.22
Metallized CNT	12.6 ± 1.2	0.498 ± 0.049	4.8 ± 1.2	7.46 ± 1.89
Sintered copper	6.6 ± 2.3	1.07 ± 0.38	0.4 ± 0.1	88.6 ± 22.2
Sintered copper + metallized CNT	0.8 ± 0.2	8.27 ± 2.07	0.5 ± 0.2	79.2 ± 31.7

^aThermophysical property values estimated from 3M Corporation [45].

^bThermophysical property values obtained from Lemmon et al. [46].

1. The application of carbon nanotubes to a flat surface reduces boiling incipience temperatures, significantly reducing the temperature excursion common for a highly wetting liquid.
2. The comparatively large-scale sintered coating is more effective than carbon nanotubes at reducing the incipience temperature.
3. The hybrid surface virtually eliminated the incipience temperature for the highly wetting HFE-7300; however, in water there was no apparent incipience benefit for the hybrid surface over the conventional sintered surface, largely because the incipience superheat was less than 1 K for both.

Previous authors [31–35] considered mechanisms by which noncondensable gas could be trapped on a surface, thereby providing two-phase interfaces for heterogeneous nucleation. Hsu’s [1] nucleation theory requires a finite wall superheat for the incipience of bubble growth to occur according to the limiting thermal boundary layer thickness in the superheated liquid. Afgan et al. [5] suggested that for thermally conductive porous heated layers, Hsu’s maximum cavity radius criterion is not applicable, because the superheated fluid within the matrix is likely to be surrounded by the heated solid at T_w . We note, however, that the boundary layer thickness is quite large (on the order of 1 mm) for the single-phase natural convection condition experienced prior to boiling incipience in the present work. In this case, only Hsu’s minimum radius criterion is important. As shown in Table 1, the minimum bubble radius depended on the pore size of the surface as well as on the wettability of each fluid.

Highly wetting liquids require low-angle cavities for gas trapping to occur (see, e.g., Reed and Mudawar [36]). Ujereh et al. [18] suggested that carbon nanotube arrays provide effective cavity angles near zero. The sintered copper surfaces both with and without CNTs in the present work, however, had incipience superheats much lower than the CNT-enhanced flat surface. We therefore conclude that the gas- or vapor-trapping capability of a sintered copper layer produces larger bubble embryos than CNTs alone. Addition of CNTs to the sintered layer, however, allows embryos eight times larger to be trapped for the highly wetting HFE-7300. This is not the case for water, for which the maximum trapped bubble size in the sintered layer already approaches the pore length scale.

Boiling Hysteresis

Three main types of hysteresis were present in the boiling curves: incipient boiling hysteresis, long time constant hysteresis, and apparent hysteresis from contamination. First,